

The Hong Kong Polytechnic University
COMP3423 Human Computer Interaction

Group Project: Designing a New Interface for an Everyday Application

(Semester 1 | Duration: 8 weeks | Due date: **15th Nov 2026 (Sun), 23:59**)

1. Project Overview

In this group project, you will work in groups of five to design a new interface for an everyday application (e.g., food delivery, mobile banking, public transit, health/fitness, smart home, e-commerce, social, or education). Each group member will assume one specific role, contributing to the project through their unique expertise. The goal is to analyze, critique, and redesign the chosen application based on Human-Computer Interaction (HCI) principles. This project emphasizes the practical application of HCI theories and principles in a collaborative environment. Your ability to effectively redesign an interface, evaluate it with real users, and communicate your findings will be essential for achieving a high grade.

The project has three stages:

Stage 1 — Analyse: identify usability problems in the existing application (accessibility barriers are a bonus item).

Stage 2 — Design: design a new interface. The interface is not limited to a phone screen: it may target a mobile app, desktop GUI, voice assistant, gesture, projection (e.g., interactive table), AR/VR, wearable, smart speaker, robot, or a multimodal combination.

Stage 3 — Prototype & Evaluate: build a Figma prototype of at least 5 major features ($N = 5$ = number of team members), test it with at least 5 real participants via Maze, and report the method, results, and discussion.

Note on non-screen interfaces: Figma is screen-based. If your team chooses a non-screen modality (e.g., robot, projection, voice), the Figma prototype must cover the user-facing screens/flows (e.g., a companion app or control panel), and the report must explicitly explain how the physical/gestural/voice layer maps onto the prototype.

2. Group Roles

Each member holds exactly one primary role from the list below (supporting other tasks is allowed). The role and key tasks must be declared in the cover-page workload table (see Appendix I).

- 1. Project Lead & UX Strategist:** Timeline and coordination; owns the Introduction and Conclusion sections of the report; ensures every design decision is explicitly tied to an HCI principle; final QA and proofread.
- 2. User Researcher:** Defines the target user group; builds the user journey; designs the Maze test (tasks, metrics, recruitment); owns the Testing section of the report.
- 3. Interaction Designer:** Information architecture, wireframes, Figma prototype structure (navigation, interactions, states); owns the application walkthrough.
- 4. Visual & UI Designer:** Visual language and style guide; high-fidelity screens in Figma; owns PPT design.
- 5. Accessibility & Evaluation Lead:** Heuristic evaluation of the existing app; accessibility audit (e.g., contrast, touch targets, text scaling, screen-reader compatibility); accessibility accommodations in the redesign.

3. Project Requirements

3.1 Task Overview

- 1. Select an Everyday Application:** Choose an everyday application (e.g., food delivery, mobile banking, public transit, health/fitness, smart home, e-commerce, social, or education). Describe its purpose, usage, and target user group.
- 2. Analyse:** Document the existing application with screen captures and identify its usability problems, explaining why each is poorly designed based on HCI theories and design principles (accessibility barriers are a bonus item).
- 3. Design:** Propose a new interface addressing the identified problems. The interface is not limited to a phone screen — see Stage 2 in the Project Overview.
- 4. Prototyping:** Create a Figma prototype demonstrating at least 5 major features (N = team size), clickable, with navigation, states, and edge cases handled.
- 5. User Testing:** Conduct a user test with at least 5 real participants using Maze to gather feedback (e.g., task success, time on task, errors, subjective rating).

3.2 Documentation and Presentation

- 1. Report (max. 20 pages, excluding references and appendices),** include the following content:

First Page or Cover Page (Workload Declaration): Project title; all team members' names & SIDs; workload declaration table (Name, SID, Role, Key tasks) with the group leader marked.

1. Introduction (~1 page): The chosen everyday application and its context; target user group; problem statement (why the existing interface fails them); scope: the 5 features to be redesigned.

2. Related Work (2–4 pages): 10–20 references critically synthesising HCI theory, studies on the chosen domain, and work on the chosen interface modality. Every reference must be cited in the text and used to justify a design decision.

3. Overall Design (3–5 pages): Application overview and purpose; an illustrated user journey for the target user group (stages, touchpoints, pain points); a walkthrough of the application with screen captures; overall design rationale mapped to HCI principles; accessibility considerations (if any).

4. Feature Details (4–6 pages): For each of the 5 features: (a) the usability problem in the existing app shown with an annotated screen capture; (b) the HCI principle(s) violated/applied, named explicitly; (c) the redesign with before/after screen captures; (d) design rationale.

5. Testing Method, Results and Discussion (3–5 pages): Maze method (participant profile with $n \geq 5$, recruitment, tasks, metrics, procedure); results with tables/figures and verbatim participant quotes; discussion interpreting results against the design decisions; iteration evidence showing what changed in the prototype after testing.

6. Conclusion and Future Work (~1 page): Summary of effort and findings consistent with the report's evidence; one concrete, justified future development direction; honest reflection on limitations.

References: 10–20, consistent citation style.

Appendices: Raw Maze data (export/dashboard screenshot), task script, Figma URL, extra screen captures.

Screen Captures: Include images of the existing application and the redesigned version. Please attach the Figma URL, **publicly accessible** (view/comment) by markers for testing and grading.

2. Presentation (PPT + presentation video):

Video: maximum 10 minutes presenting the work. Suggested structure: problem & users (1.5 min) → user journey (1 min) → design & 5 features with live Figma walkthrough (4 min) → Maze testing & results (2.5 min) → conclusion & future direction (1 min). The Figma prototype must be demonstrated on screen.

PPT: clean, readable, screenshot-driven (no walls of text).

4. Suggested Timeline (8 Weeks)

Week	Dates (indicative)	Milestone
1	22–28 Sep	Kick-off: form groups, confirm the 5 roles, choose the everyday application and target user group.
2	29 Sep–5 Oct	Usability analysis of the existing app (annotated screen captures); start related-work reading; define target users.
3	6–12 Oct	User journey + personas; information architecture; low-fidelity wireframes; finalise the list of 5 major features.
4	13–19 Oct	Mandatory mid-project checkpoint (ungraded gate): submit a 1–2 page progress snapshot (app choice, roles, user journey draft, wireframes, feature list). High-fidelity Figma work on features 1–3.
5	20–26 Oct	Complete Figma prototype of all 5 features; finalise Maze test plan (tasks, metrics, recruitment).
6	27 Oct–2 Nov	Run the Maze test with ≥ 5 participants; collect raw data.
7	3–9 Nov	Analyse results; iterate the prototype based on findings; write the report; draft PPT.
8	10–15 Nov	Finalise report + PPT; record the ≤ 10 -min video; submit everything by 15th Nov (Sun), 23:59.

5. Assessment Criteria (Total 100%, 5% Accessibility Bonus)

Every rubric item is a **checkable evidence requirement** (yes/no), and marks are awarded pro-rata for the items met. Claims without supporting evidence score 0 for that item — this rewards genuine process work and keeps grading fair and objective. The full rubric with checkable items is in Appendix II.

Part A: Report (86%)

Introduction & Problem Framing (8%) — problem statement, target user group, justification of app choice, realistic scope of the 5 features.

Related Work (12%) — 10–20 references, all cited; balanced mix of HCI theory, domain, and modality; critical synthesis; design decisions traceable to the literature.

Overall Design (18%) — application overview; illustrated user journey with pain points; walkthrough with screen captures; rationale tied to named HCI principles; overall coherence and consistency.

Feature Details (20%) — all 5 features prototyped in Figma and clickable via the public link; annotated problem evidence from the real existing app; named HCI principles; before/after captures; prototype quality (navigation, states, edge cases).

Testing Method, Results and Discussion (20%) — Maze test actually run with ≥ 5 real participants with raw data; method fully described; results with data and verbatim quotes; discussion linked to design decisions; iteration evidence.

Conclusion & Future Work (8%) — summary consistent with the report's evidence; one concrete justified future direction; honest limitations.

Part B: Presentation & Video (14%)

Video ≤ 10 min covering all required content; clear, professional delivery; PPT quality (readable, screenshot-driven); Figma prototype demonstrated in the video.

Accessibility Bonus (+5%, capped at 100%)

Documented accessibility audit of the existing app (2%); accessibility accommodations in the redesign, shown and explained (2%); accessibility considered in the Maze test (1%).

Submission Guidelines:

Submit one zip file per group on Canvas containing the report (MS Word or PDF) and the presentation PPT.

Each group only needs to submit to Canvas ONCE by a representative or group leader.

The presentation video is uploaded separately to the link provided on Canvas (not inside the zip).

Marks will be deducted if the markers are unable to access the Figma link.

Please note the late penalty policy for the HCI course.

Appendix I: Example Cover Page

COMP3423 Human Computer Interaction — Group Project Report

Project Title: Re-design the XXXX app (new interface for ...)

Group XX

Workload declaration table

Name	SID	Role	Key tasks
*Chan Tai Man	S1234567d	Project Lead & UX Strategist	Timeline, Introduction & Conclusion, HCI-principle mapping, QA
John Baker	S2341234x	User Researcher	User journey, Maze test design, Testing section
Happy Chan	S2341232d	Interaction Designer	IA, wireframes, Figma prototype, walkthrough
...	...	Visual & UI Designer	Style guide, high-fidelity screens, PPT
...	...	Accessibility & Evaluation Lead	Heuristic + accessibility audit, accommodations

*Group leader

Note:

1. The above name/sid/roles/tasks are examples only. You need to fill in your own information.
2. Put this form on the first/cover page of the report.
3. The key tasks do not need to be too long; they should be as precise as possible. Please refer to the corresponding section or figures in the report.
4. You don't need to sign the declaration form. Once submitted, we assume that every team member agreed on the report's content and the workload declaration form.

Appendix II: The Rubrics (100% + 5% Accessibility Bonus)

Criteria	Checkable items (marks)
1. Introduction & Problem Framing (8%)	Clear problem statement + defined target user group (2) · Justification of app choice and why the existing interface fails (2) · Scope clearly states the 5 features (2) · Scope realistic for 5 students in 8 weeks (2)
2. Related Work (12%)	10–20 references, all cited in the body (3) · Balanced mix: HCI theory + domain + interface modality (3) · Critical synthesis, not a list of summaries (3) · Design decisions traceable to the literature (3)
3. Overall Design (18%)	Application overview and purpose clearly stated (3) · User journey illustrated for the target user group, with pain points (4) · Walkthrough with screen captures, easy to follow (4) · Rationale explicitly tied to named HCI principles (4) · Interface coherent and consistent as a whole (3)
4. Feature Details (20%)	All 5 features prototyped in Figma and clickable via the public link (4) · Each feature shows the problem with annotated evidence from the real existing app (4) · Each feature names the HCI principle(s) applied (3) · Before/after screen captures for every feature (4) · Prototype quality: navigation, states, edge cases handled (5)
5. Testing Method, Results and Discussion (20%)	Maze test actually run with ≥ 5 real participants; raw data + dashboard screenshot in appendix (5) · Method fully described: recruitment, tasks, metrics, procedure (4) · Results presented with data (tables/figures) and verbatim quotes (4) · Discussion interprets results and links them to design decisions (4) · Iteration evidence: prototype visibly changed in response to test findings (3)
6. Conclusion & Future Work (8%)	Summary of effort/findings consistent with the report's evidence (3) · One concrete, justified future development direction (3) · Honest discussion of limitations (2)
7. Presentation & Video (14%)	Video ≤ 10 min covering all required content (3) · Clear, professional delivery; audience can follow (4) · PPT quality: readable, screenshot-driven (4) · Figma prototype demonstrated in the video (3)
Accessibility Bonus (+5%, total capped at 105%)	Documented accessibility audit of the existing app (e.g., WCAG basics: contrast, touch target size, text scaling, screen-reader

	compatibility) (2) · Accessibility accommodations in the redesign, shown and explained (2) · Accessibility considered in the Maze test (e.g., accessibility-focused task or diverse participants) (1)
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Marker notes — verifying genuine effort

- Maze: check the raw export (timestamps, per-participant, per-task metrics) and that quotes are specific and natural; a generic or implausible data table should trigger a query.
- "Before" screenshots must be from the real existing app, not AI-generated mockups.
- Iteration: look for dated v1 vs. v2 comparisons (Figma version history is acceptable evidence).
- Workload declaration: each declared role should have a visible, distinct contribution in the report; a member with no traceable contribution should be flagged.
- References: spot-check 2–3 references for real existence and correct usage.
- Where an item lacks evidence, award 0 for that item — do not award marks for plausibility.

- End of Project Description -